

Start Date: 11/21/2024

Music Controller System
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Using Java

Figure 1.1 - Overview of the Music Controller System

- This system will control music playback from a variety of sources: Spotify, YouTube music, Apple music, Amazon music, SoundCloud, iHeartRadio, and Pandora. This can be done through APIs, a web browser set up, or the native applications ✓
- The system will need to be able to create playlists within the app for each music source for the user to access and play from ✓
- The app will give the user the option to download the music to the device for offline listening ✓
- The app should control playback with different speakers plugged into the device ✓
- The app should have a landing page to select the different music sources ✓
- The app should allow custom profiles to be created for each music source (ie. landing page with different accounts that save the login info for the user) ✓
- The app should have a guest profile for general listening that will not save the log in info from the previous user ✓
- The app should pull playlist information from any account signed in and save it to the user's folder ✓
- The app should have the Music Controller Logo displayed somewhere in the app ✓
- There should be multiple navigation options on the left-hand side of the app ✓
- There should be a settings menu to change the appearance of the app ✓
- The system should not take any longer than 30 seconds for the user to log in and access their music ✓
- The system should not take any longer than 1 minute to log in to an account ✓
- After 30 minutes of inactivity, the app should go back to the "Choose User" landing page ✓
- The app should include images of the different music sources listed above ✓
- The app should offer a description of each music source and how it is different from the other sources ✓
- The app should display the amount of time completed in a song and the amount of time total for the song ✓
- The app should allow the user to skip a song and play the next song, or a random song from the playlist they are on ✓
- The app should show the time and date in the top-right hand corner of the screen, as well as the temperature and weather in the top-middle of the screen ✓
- Allow the user to create a queue of songs to play ✓
- Allow the user to play music in certain rooms and not in others ✓

- Possibly allow the user to play different songs in different rooms ✓
- Auto-skip ads that may pop up ✓
- Needs to run as a native app on windows 10 or 11 ✓
- The system will require the user to create a password when saving a user profile ✓

Figure 2.1 - Nonfunctional and Functional Requirements

Nonfunctional

1. Operational Requirements

- a. The System will run in a Windows environment
- b. The System will need a landing page to select different music sources
- c. The System will need a landing page to select a user profile
 - i. The system will display a Guest user on this landing page
- d. The System will pull the user's playlists from their account and back it up
- e. The System will organize the different functions with a navigation panel on the left-hand side of the app
- f. The System should allow the user to change the appearance of the app
- g. The System should have an area at the bottom that describes the system and how it works
- h. The System should have an area at the bottom that allows the user to send feedback or ask for support

2. Performance Requirements

- a. It should take less than 30 seconds for the user to log in and access their music
- b. It should take less than 1 minute for the system to pull saved log in information for the user

3. Security Requirements

- a. The system should go back to log in landing page after 30 minutes of inactivity
- b. Require the user to set a password and log in to their user account with the app

4. Cultural and Political Requirements

- a. Display the Music Controller logo in the app
- b. Display the logos of the different music sources (ie. Spotify, YouTube, etc.)
- c. Display a description of each music source and how it is different from other music sources (ie. Ads, number of songs, etc.)

Functional

1. Control Music Playback
 - a. Control YouTube music, Spotify, Apple music, Amazon music, SoundCloud, iHeartRadio, and Pandora playback
 - i. Control the speakers used in playback
 1. Select which speakers will be used when playing song
 2. Select the volume for the speakers individually or in total
 3. Select different songs for different speakers
 - ii. Display time left, and length of a song (ie. 2 mins 30 sec of 3 mins 25 sec played)
 - iii. Create Queue for music playback from different and same music sources
 1. Allow user to add multiple songs or one song
 - a. If only one song is added, allow the user to choose to continue with songs similar, or to quit playing music
 2. Keep track of which source the music comes from
 - iv. Auto-skip ads that show up on music
2. Create Playlists
 - a. Create a folder labeled with the specific label given by the user
 - b. Allow the user to add or remove songs from the specific playlist folder
 - c. Allow the user to add this playlist to their specific music resource
3. Download Music
 - a. Run the specific link through a music installer and save it to a downloaded music folder
 - b. Show the downloaded music folder as an option in the landing page
4. Display time, date, and weather for the current location
 - a. Ask the user for their area code
 - b. Find location and information from the given area code
 - c. Display time, data, and weather in the top-middle of the application

Figure 3.1 - Use Case Diagram

https://lucid.app/lucidchart/2a423472-1321-47ff-98b5-127227fd27d0/edit?beaconFlowId=E2D0E0EFAE51C993&invitationId=inv_99223609-04ac-47bb-99c4-9c83eca1bd95&page=.Q4MUjXso07N#

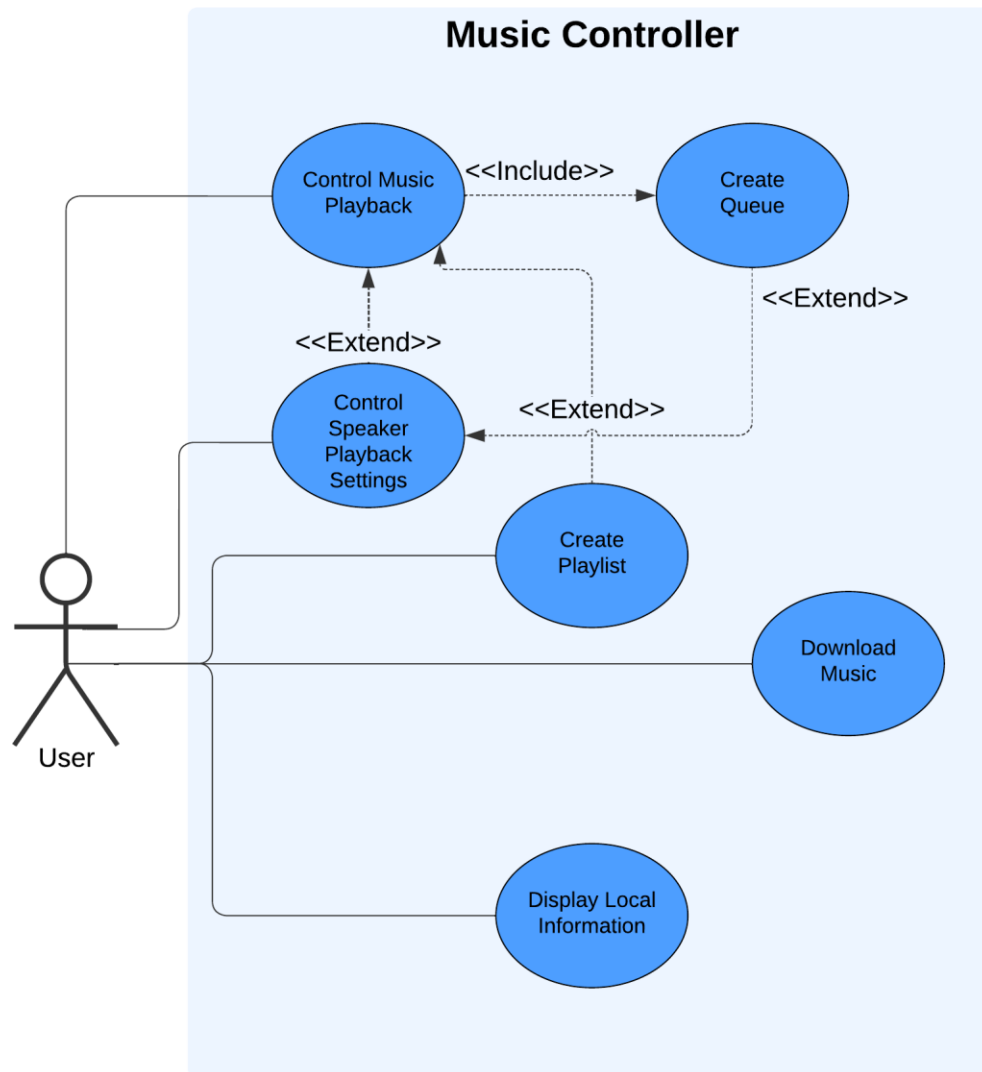


Figure 4.1 - Use Case Description: Control Music Playback

Use Case Name: Control Music Playback		ID: 1	Importance Level: High
Primary Actor: User	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case will allow the user to control what music is played and from what source the music is played from.			
Trigger: User chooses music source Type:			
Relationships: Association: User Include: Create Queue Extend: Generalization:			
Normal Flow of Events: 1. The user logs in with their specified account 2. The user chooses to play music from a specific source or multiple sources 3. The user chooses a specific song, songs, playlist, or playlists to play 4. The user does not set up playback settings for multiple devices 5. The queue and number of queues pop up in the far right-hand side of the screen 6. The user can continue to play, pause, skip, or restart songs 7. Ads are automatically skipped when able to			
SubFlows: S-1:			
Alternate/Exceptional Flows: 1a. The user does not have an account. 1b. The user is prompted to create an account with a password and username or continue as guest 1c. The user can log in to each music source with their personal accounts 4a. The user chooses to change playback settings to play multiple songs in different areas 4b. Multiple queues are set up for each area created			

Figure 4.2 - Use Case Description: Create Queue

Use Case Name: Create Queue		ID: 2	Importance Level: High
Primary Actor: System	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case description will create a queue based on what the user decides to play through the app or will create multiple queues depending on how many speakers are playing different songs.			
Trigger: User picks a song or a playlist to play, or the user sets different songs for different speakers. Type:			
Relationships: Association: User Include: Extend: Control Speaker Playback Settings Generalization:			
Normal Flow of Events: 1. The user chooses a song, multiple songs, a playlist, or multiple playlists to play on selected devices 2. The song information is pulled and is set up in a queue that will list each song in a FIFO method 3. Each song will be played in accordance with the queue pulling the necessary information from whatever source the song comes from			
SubFlows: S-1:			
Alternate/Exceptional Flows: 1a. The user chooses multiple songs or playlists to play in different areas. 1b. Multiple queues are set up in accordance with the different areas and the songs or playlists selected by the user.			

Figure 4.3 - Use Case Description: Control Speaker Playback Settings

Use Case Name: Control Speaker Playback Settings		ID: 3	Importance Level: High
Primary Actor: User	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case description allows the user to specify which speakers are playing what songs. If more than one speaker is playing different songs, then multiple queues are created.			
Trigger: The client would like to modify the settings for speaker playback. Type:			
Relationships: Association: User Include: Extend: Control User Playback Generalization:			
Normal Flow of Events: 1. The user chooses to modify the speaker playback settings 2. The system displays the different speakers and their areas for the user to choose to play certain music in 3. The user changes the specific volume for each speaker in their respective areas 4. The user selects save to save their current configuration 5. After the user logs out of their account, the Speakers Playback Config is reset			
SubFlows: S-1:			
Alternate/Exceptional Flows: 1a. The user chooses not to modify the speaker playback settings 1b. Set default settings that will play the music to every device at half volume 1aa. The user selects a saved config. 4a. The user chooses to not save the config 4b. The config is reset back to default 5a. The user chooses to save this config as a config option 5b. Save this config and allow the user to choose this config when setting up speaker playback settings			

Figure 4.4 - Use Case Description: Create Playlist

Use Case Name: Create Playlist		ID: 4	Importance Level: Medium
Primary Actor: User	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case description will allow the user to create a local playlist that will contain songs from any music source they would like. This allows the user to start this playlist and play different songs from different sources.			
Trigger: When the user starts playing music they can add the song to a playlist, or the user can go through and pick songs to add to a specified playlist.			
Type:			
Relationships: Association: User Include: Extend: Control Music Playback Generalization:			
Normal Flow of Events: 1. The user would like to add songs to a playlist 2. The user adds songs or playlists one by one or selects multiple songs or playlists to add to the playlist 3. The songs are added to a local folder with all the information and data related to the song (including the source it came from)			
SubFlows: S-1:			
Alternate/Exceptional Flows:			

Figure 4.5 - Use Case Description: Download Music

Use Case Name: Download Music		ID: 5	Importance Level: Low
Primary Actor: User	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case description allows the user to pick certain songs to download from any music source they would like and store the music locally.			
Trigger: The user decides to download music from a specified source Type:			
Relationships: Association: User Include: Extend: Generalization:			
Normal Flow of Events: 1. The user decides to download music from a certain source 2. The user can download songs or playlists one by one or select multiple songs or playlists to download 3. These songs are downloaded in mp3 format and saved to a local folder on the computer 4. The user can then access these songs or playlists when offline			
SubFlows: S-1:			
Alternate/Exceptional Flows:			

Figure 4.6 - Use Case Description: Display Local Information

Use Case Name: Display Local Information		ID: 6	Importance Level: Low
Primary Actor: User	Use Case Type:		
Stakeholders and Interests:			
Brief Description: This use case description allows the user to enter their area code, which will then display the current weather, time, and date.			
Trigger: User decides to enter their area code to display current weather, time, and date. Type:			
Relationships: Association: User Include: Extend: Generalization:			
Normal Flow of Events: 1. The user would like local information to be displayed on the screen 2. The user goes to their settings to enter their zip code 3. The System will then look up the zip code and bring in information about the weather, time and date at that location			
SubFlows: S-1:			
Alternate/Exceptional Flows:			

Figure 5.1 - Activity Diagram: Control Music Playback

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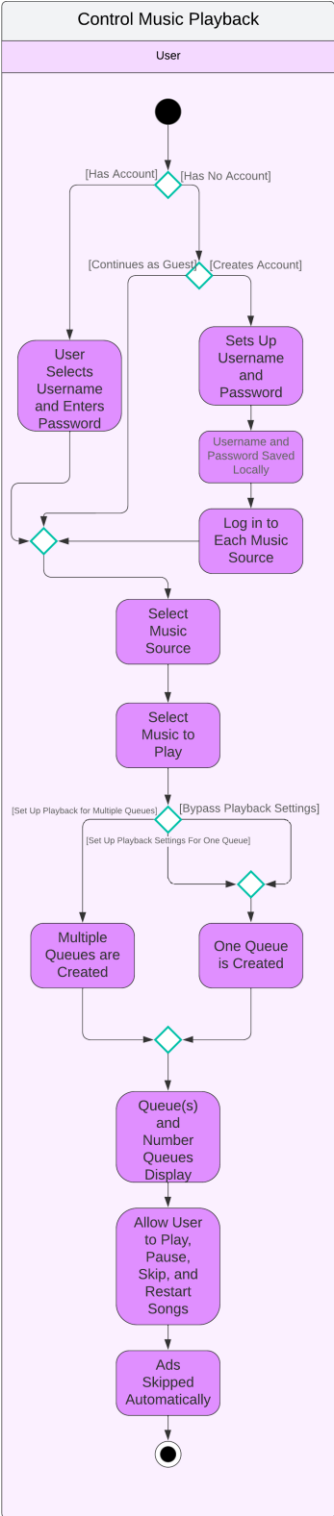


Figure 5.2 - Activity Diagram: Create Queue

https://lucid.app/lucidchart/33e402f7-c246-4e82-8e69-4739654c64cb/edit?invitationId=inv_8aef4885-0676-4c1d-aa53-6af263558b69&page=5RCtcjNsiEMH#

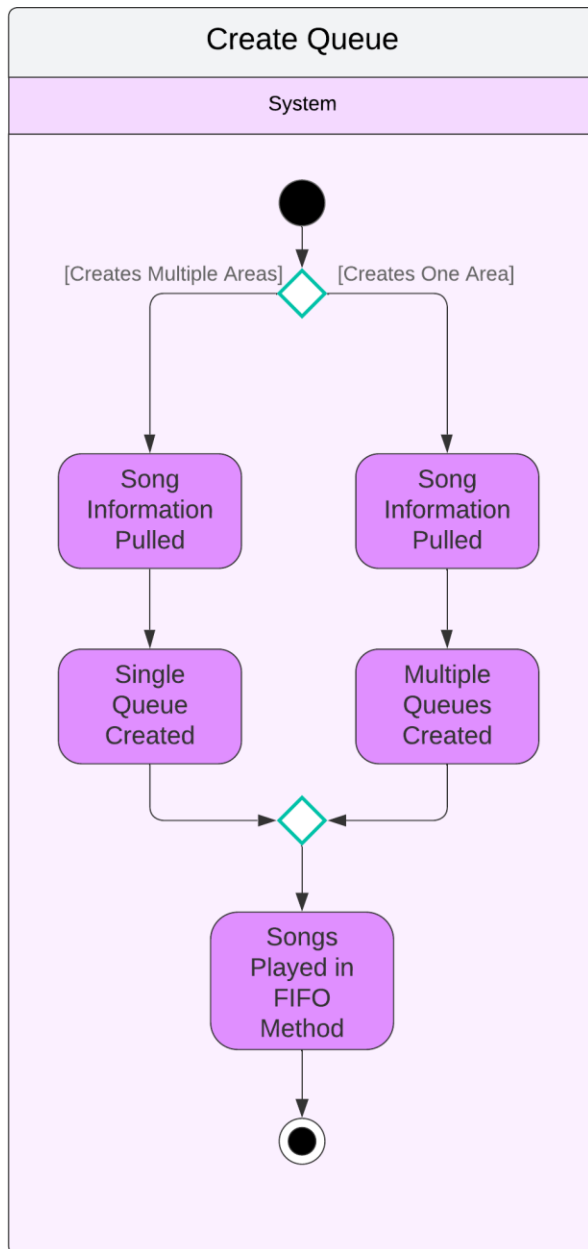


Figure 5.3 - Activity Diagram: Control Speaker Playback Settings

https://lucid.app/lucidchart/33e402f7-c246-4e82-8e69-4739654c64cb/edit?invitationId=inv_8aef4885-0676-4c1d-aa53-6af263558b69&page=dSCt9ONgSDGF#

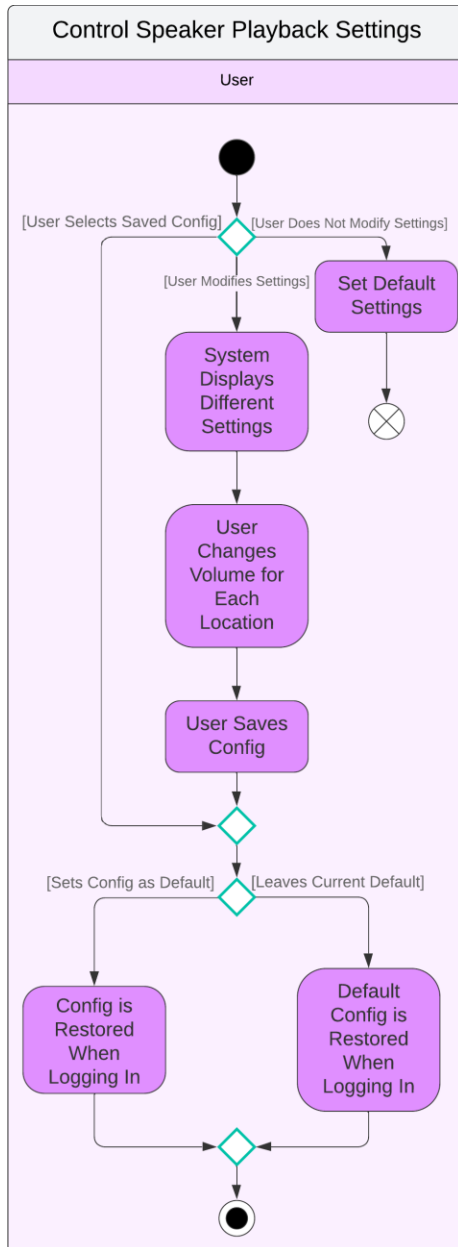


Figure 5.4 - Activity Diagram: Create Playlist

https://lucid.app/lucidchart/33e402f7-c246-4e82-8e69-4739654c64cb/edit?invitationId=inv_8aef4885-0676-4c1d-aa53-6af263558b69&page=wSCtGX07P8ab#

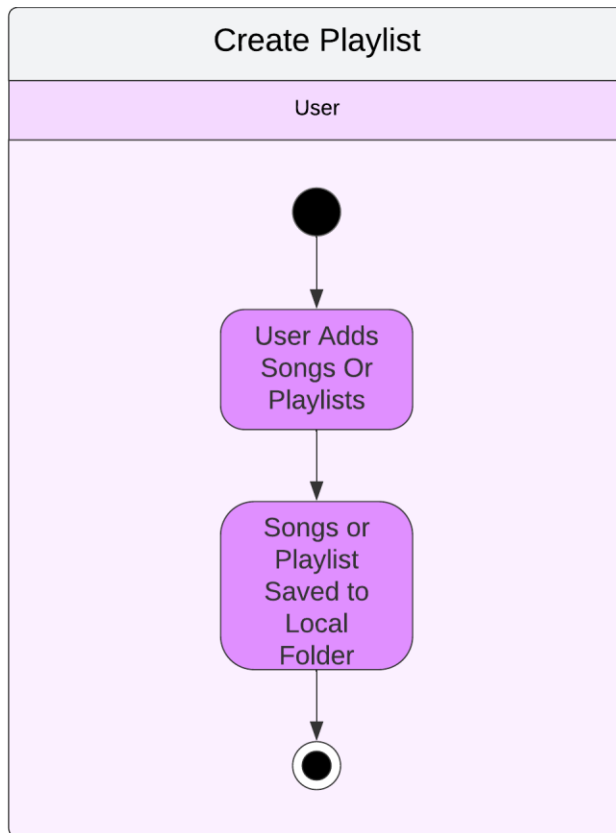


Figure 5.5 - Activity Diagram: Download Music

https://lucid.app/lucidchart/33e402f7-c246-4e82-8e69-4739654c64cb/edit?invitationId=inv_8aef4885-0676-4c1d-aa53-6af263558b69&page=FSCT6uGwEido#

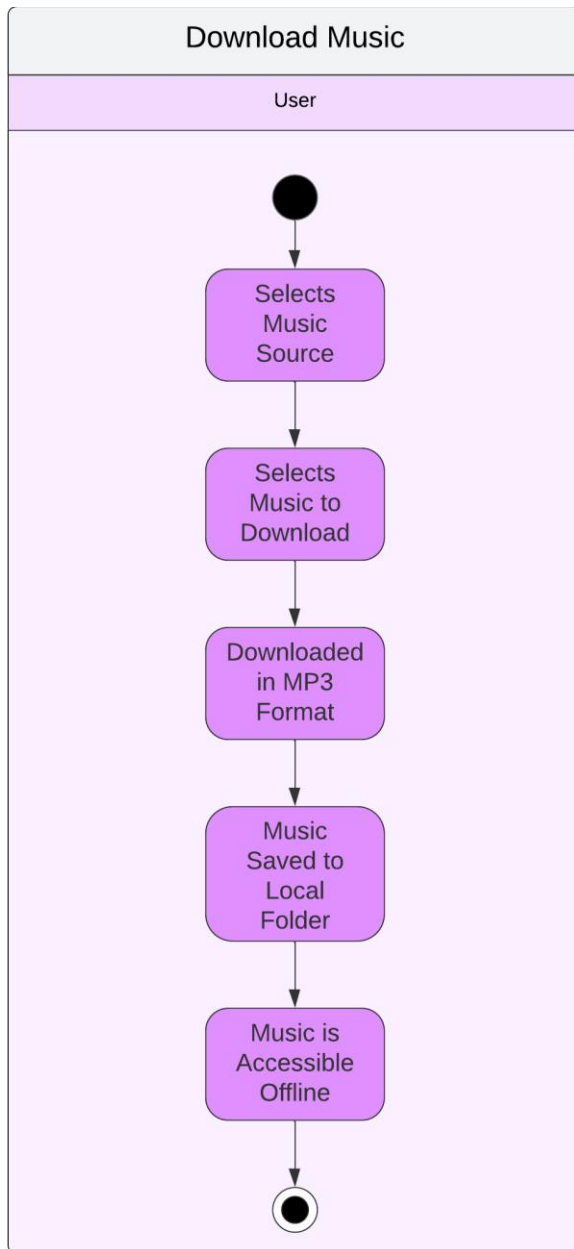


Figure 5.6 - Activity Diagram: Display Local Information

https://lucid.app/lucidchart/33e402f7-c246-4e82-8e69-4739654c64cb/edit?invitationId=inv_8aef4885-0676-4c1d-aa53-6af263558b69&page=QSCtCtU_M2Kz#

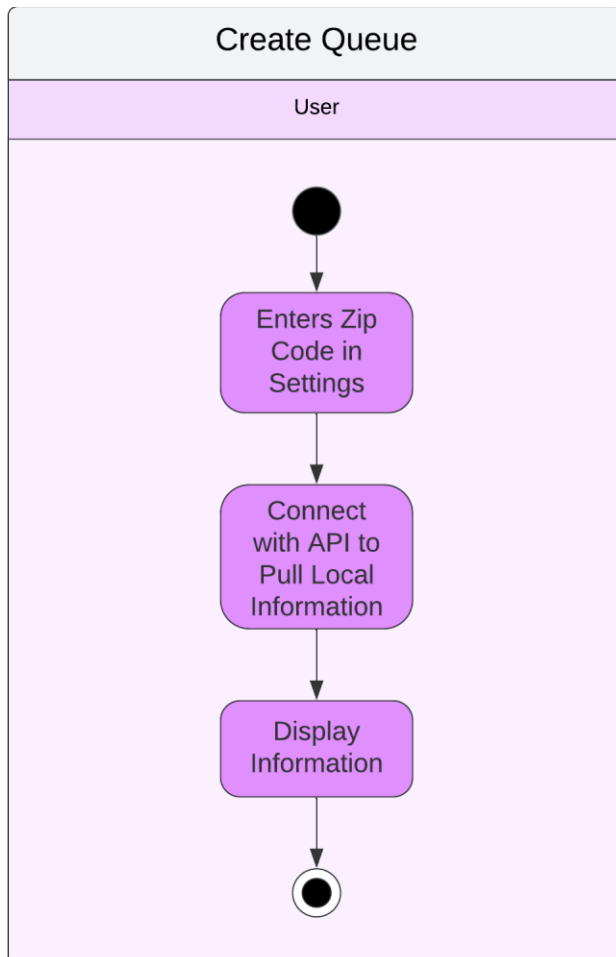


Figure 6.1 - Working Class Tables: Grammatical Objects

Grammatical Objects	Class	Comments
Account	√	Class Account
User		Actor
Music		Attribute (Music Sources)
Music Sources	√	Class Music Sources
Song		Attribute (Music Sources)
Playlist		Attribute (Account)
Playback Settings		Attribute (Account)
Speakers		Attribute (Account)
Queue	√	Class Queue
Number of Queues		Attribute (Queue)
Play		Attribute (Queue)
Pause		Attribute (Queue)
Skip		Attribute (Queue)
Restart		Attribute (Queue)
Account Password		Attribute (Account)
Account Username		Attribute (Account)
Guest		Attribute (Account)
Music Accounts		Attribute (Account)
Areas		Attribute (Account)
Song Information		Attribute (Music Sources)
System		Actor
Volume		Attribute (Account)
Current Configuration		Attribute (Account)
Default Config		Attribute (Account)
Saved Config		Attribute (Account)
Downloaded Music		Attribute (Account)
Computer		Actor
Local Information	√	Class Local Information
Zip Code		Attribute (Local Information)
Weather		Attribute (Local Information)
Time		Attribute (Local Information)
Date		Attribute (Local Information)
Location		Attribute (Local Information)

Figure 6.2 - Working Class Tables: Class Descriptions

Class	Description
Account	Contains all personal user settings, accounts, playlists, playback settings, speakers, account password, account username, guest account, music accounts, speaker areas, speaker volume, speaker current config, speaker default config, saved config, and downloaded music. This is where the user will create a log in and sign in to their music sources within their account. This is also where the user can modify all settings regarding sound and personal information.
Music Sources	Contains music, song, and song information. This class will pull information from different music sources and show the users recommended songs and allow the user to play certain songs.
Queue	Contains the number of queues, the ability to play, pause, skip, and restart. This controls the music playback for the user; the user can control how music is played and where the music is played to.
Local Information	Contains zip code, weather, time, date, and location. This class allows the user to enter their zip code, which will then pull all local information to display in the app.

Figure 6.3 - Working Class Tables: Class Responsibilities and Use Cases

Class	Responsibilities	Use Cases
Account	Account Username Account Password Default Playback Settings Current Playback Settings Account Playlists Number of Speakers Saved Configs Downloaded Music Different Accounts Music Accounts Guest Account Speaker Areas Speaker Volume	Control Music Playback
Music Sources	Music Song Song Information Playlists	Control Music Playback Create Queue Download Music
Queue	Number of Queues Play Pause Skip Restart	Control Music Playback Create Queue
Local Information	Zip Code Weather Time Date Location	Display Local Information

Figure 7.1 - Class Diagram

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